Weekly Review
Brainscape Test 2

Complete the weekly review, you'll have 15 minutes.

Weekly Review
Brainscape Test 2

Mark that review!

Once you've done it:

1. Submit the score into the Google Form.
2. Update your weekly review tracker with **targets**.

Try to decode:

```
20,15,4,1,25   23,5   1,18,5  12,5,1,18,14,9,14,7
1,2,15,21,20   19,5,3,18,5,20   3,15,4,5,19    1,14,4
3,15,13,16,21,20,5,18    19,3,9,5,14,3,5    9,19
7,18,5,1,20!
```

Try to decode:

```
20,15,4,1,25   23,5   1,18,5  12,5,1,18,14,9,14,7
1,2,15,21,20   19,5,3,18,5,20   3,15,4,5,19    1,14,4
3,15,13,16,21,20,5,18    19,3,9,5,14,3,5    9,19
7,18,5,1,20!
```

Today we are learning about secret codes and computer science is great!

Symmetric Encryption

Keywords: Encryption; Decryption; Cipher; Plaintext; Ciphertext; Caesar Cipher; Vernam Cipher; One-Time Pad; Mathematical Security; Computational Security

Practice with Encryption

The best way to get to grips with this is to see what you've already got and can work out!

Practice with Encryption

The best way to get to grips with this is to see what you've already got and can work out!

A murder has been committed at Barton Peveril.
Your job, in pairs, is to decode the clues to find

- the identity of the murderer
- the murder weapon
- the room in which the murder took place

When you have the answer to a clue, write it as a private comment. If correct, you'll gain access to the next clue.
All 5 clues are needed to solve the murder.
YOU ARE NOT TO USE THE INTERNET TO HELP YOU EXCEPT WHEN THE CLUE TELLS YOU!!!!

Terminology

Encryption

Cipher

Plaintext

Ciphertext

Terminology

Encryption

- Using an algorithm to change data so that it can only be read by the intended recipient(s)

Cipher

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Terminology

Encryption

- Using an algorithm to change data so that it can only be read by the intended recipient(s)

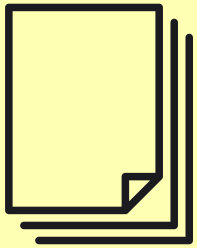
Cipher

- The algorithm which is used to encrypt or decrypt

Plaintext

Ciphertext

Terminology



Encryption

- Using an algorithm to change data so that it can only be read by the intended recipient(s)

Cipher

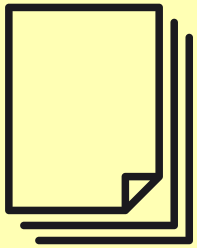
- The algorithm which is used to encrypt or decrypt

Plaintext

- The unencrypted message

Ciphertext

Terminology



Encryption

- Using an algorithm to change data so that it can only be read by the intended recipient(s)

Cipher

- The algorithm which is used to encrypt or decrypt

Plaintext

- The unencrypted message

Ciphertext

- The encrypted message

Symmetric Encryption

With a **symmetric** encryption scheme, the **same key**, k, is used for both the **encryption and decryption** algorithms.

In **asymmetric** encryption, **different** keys are used.

Caesar Cipher

The Caesar cipher is a type of **substitution cipher**. It replaces each plaintext letter with a different one a fixed number of places down the alphabet.

eg
Replace each letter L with (L+3) mod 26
“Attack at dawn” becomes “Dwwdfn dw gdzq”

Two components:

- Algorithm: Shift characters by a fixed amount
- Key: the fixed amount each letter is shifted

You Try

1. Encrypt “Hello, world” using a key of 4.

2. Decrypt “Tevsec Mkockb” using a key of 10.

3. **Extension:** Can you crack the ciphertext
   “alex mw xlmw xcti sj irgvctxmsr geppih”

You Try

1. Encrypt “Hello, world” using a key of 4.
   “Lipps Asvph”
2. Decrypt “Tevsec Mkockb” using a key of 10.
   “Julius Caesar”
3. **Extension:** Can you crack the ciphertext
   “alex mw xlmw xcti sj irgvctxmsr geppih”
   “what is this type of encryption called” k = 4 Answer: Substitution Cipher

Pros and Cons

Discuss in pairs: What are the disadvantages of the Caesar Cipher?
Write down two.

Pros and Cons

Discuss in pairs: What are the disadvantages of the Caesar Cipher?
Write down two.

- Word structure is preserved.
  - Solution: break message into equal-length blocks.
    - dww dfn dwg dzq
- Letter frequency is a big clue
  - e,t,a,o most common English letters.
  - Using a single key preserves frequency.
- Very quick to crack using brute force
  - Fundamental problem: key length (in binary) is much shorter than the message

Vernam Cipher

AKA - One Time Pad

Concept: Bitwise XOR the whole message with an equally long random binary key.

The key must be:

- The same length as the message (in binary)
- XORed with the plaintext to prodice ciphertext

Usefully, XOR is a self-inversing function, so if both people have the key, they can both decrypt!

You Try

1. Encrypt "Hi" using a key of `1011 1001  0101 0111`.

2. **Extension**: Why is the Vernam cipher only secure if you never re-use the same key?

You Try

1. Encrypt "Hi" using a key of `1011 1001  0101 0111`.

`(int)'H' =72 =0100 1000, (int)'i' =105 =0110 1001`

Ciphertext: `1111 0001  0011 1110`

2. **Extension**: Why is the Vernam cipher only secure if you never re-use the same key?

Vernam Cipher Pros and Cons

Pros:

- No two bits are encoded with the same mapping
  - an attacker has no frequency information to help guess the key.
- Perfectly secure.
- Attacker can do no better than guessing.
- Ciphertext gives no information about plaintext.

Cons:

- Keys are very large.
- How do you distribute this key?
- Needs a shared source of randomness.

Experiment

In pairs: Ciphertext C =

`11010111 11111000 01011101 00011011 01001011` Decrypt using a key of:

- `10010100 10010000 00101111 01110010 00111000` k_A
- `10010000 10001010 00111000 01111110 00100101` k_B
- `10111010 11001011 00110000 00101000 01111110` k_C
- `10000111 10001010 00111000 01101000 00111111` k_D

Which is the correct key?

Extra: Transposition Cipher

EG: Eg Rail Fence Using a Key of 5, encrypt the message
COMPUTING IS GREAT FUN

Extra: Transposition Cipher

EG: Eg Rail Fence Using a Key of 5, encrypt the message COMPUTING IS GREAT FUN

C	O	M	P	U
T	I	N	G	I
S	G	R	E	A
T	F	U	N	

Extra: Transposition Cipher

EG: Eg Rail Fence Using a Key of 5, encrypt the message COMPUTING IS GREAT FUN

C	O	M	P	U
T	I	N	G	I
S	G	R	E	A
T	F	U	N	

CTSTOIGFMNRUPGENUIAZ

Let's practice!

On BPCompSci:

W119 - C# Encryption Programming